

Professional Ethics in Computing – Seminar 8

In today's seminar you will investigate computer failure related to a real world case. You should work individually, but may talk together to share ideas.

Here are some quick guide answers to each question.

Step 1. Read the report you have been given. Form an intuitive view of what happened. (approx. 20 minutes)

Step 2. Answer the following questions (approx. 20 minutes):

1. What kind of computer failure was involved in this case?

Mainly software error, but also some communication failure.

2. Analyse the responsibilities of the computer programmers using Birsch's three criteria.

- a. The programmers wrote code that did not work correctly and was a significant cause of the accidents.
- b. The programmers were negligent, since this was a safety-critical system and they should have ensured that rigorous testing and fail-safe design was included.
- c. The programmers knew that this was a safety-critical system and the importance of correct operation. Also, there is strong evidence that this weakness was known before the accidents happened.

3. What, if anything, could have been done to prevent the incidents?

Implement more full testing; implement redundancy feature; proper reporting to FAA; follow-up carefully with test pilot concerns, rather than ignore.

4. Who, if anyone, should take moral responsibility?

The programmers, testers and managers at Boeing.

5. Who, if anyone, should take legal responsibility?

Boeing company and Boeing executive officer overseeing airplane development.

Step 3. Sum up. (approx. 10 minutes).

See the outcome report to the case.